VIACHAIN™
AN ORBCOMM COMPANY



CT 5300
Installation Guide

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CONTACT AND LEGAL INFORMATION

Visit VIACHAIN Online

Website:	www.viachain.io
Head Office:	22970 Indian Creek Drive, Suite 300, Sterling, VA 20166 USA

Contact Customer Care

Website	https://www.orbcomm.com/en/support	
Email	customer.care@orbcomm.com OR FMSupport@orbcomm.com (Fleet Management support needs)	
Phone	(North America Toll-Free) 1.800.ORBCOMM	(United Kingdom Toll-Free) +44 800 538.5909
	(Australia Toll) +61 (8) 6186 9633	(Austria, Germany, Switzerland Toll) +49 89.208045522)
	(New Zealand Toll) +64 (9) 884 1439	(Ireland Toll) +353 1.582.4013

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1 INSTALLATION

This document covers both container and thin-walled trailer installations.



IMPORTANT

READ ALL INSTRUCTIONS CAREFULLY BEFORE INSTALLING. FAILURE TO DO SO MAY CAUSE PERSONAL INJURY OR DAMAGE TO PRODUCT AND/OR PROPERTY.

• Review the product package and contents prior to beginning the installation. Take care when opening the packaging and removing items. If a return is needed, you will want to return the product in its original packaging if possible. • This instruction guide is provided as a GENERAL installation guide; some assets vary dimensionally and may require additional steps. • ORBCOMM has a policy of continuous development and improvements. Therefore, products, guides, and technical information are subject to change without prior notice. • The manufacturer and / or distributors do not accept responsibility for third-party charges, labor, and / or third-party replacement modifications that are not ORBCOMM approved. Some modifications may void the factory warranty. • ORBCOMM does not accept any responsibility for installations performed by installers / third parties not approved and / or authorized by ORBCOMM. Some installations may void the factory warranty. • Exercise due diligence when installing this product. ORBCOMM does not accept any responsibility for asset damage or personal injury resulting from the installation of this product. Careless installation and operation can result in serious injury or equipment damage. • All liability for installation and use rests with the owner / operator. • Always make sure you have a clean, dry, and well-lit work area. • Always ensure products are secure during disassembly and installation. • Always take steps to protect yourself when drilling, cutting, and grinding because this may create flying particles that can cause injury. • Thoroughly inspect the area to be drilled, on both sides of material, prior to modification, and relocate any objects that may become damaged. • Always route electrical cables carefully. Avoid moving parts, parts that may become hot and rough, or sharp edges. • Make sure to fully understand the product, its intended use, and operation prior to use.

CAUTION: While we provide mounting hardware to assist with installations, it is the responsibility of the installer to select the proper mounting hardware for the asset's surface material where a device or accessory will be mounted.

CAUTION: Always follow local disposal guidelines to properly dispose of the Lithium-ion battery and the device.

CAUTION: DO NOT replace the battery. Changing the battery without ORBCOMM's permission could violate regulatory conformity.

CAUTION: DO NOT throw the internal battery or the device into fire.

1.1 Wake Up and Associate the Device

1. Remove the CT 5300 from the packaging and record its **serial number** (device ID) and the associated **asset identification number** (asset ID).

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2. Expose the device's solar panel to light to wake up the device. The device ships in a non-transmitting mode.

   Note: Pairing the CT 5300 with an asset is not possible without waking up the device. Expose the CT 5300 to sunlight at the beginning of the day to ensure it has time to wake up and transmit critical information before you install it.
   Alternatively, you can use another bright light source to wake up the device. Use a light meter or a mobile phone app to find a light source with a minimum 600 Lux light intensity, the higher the Lux value, the better. Position the CT 5300 solar panel directly facing the light source for at least one minute.

   CAUTION: When using a hot light source (for example, halogen or incandescent), ensure that the solar panel does not become hot to the touch.

3. Use the ORBCOMM Installer app to associate the device ID with the asset ID.

Android Version (Google Play Store)

iOS Version (Apple App Store)

1.2 Mount the CT 5300 using Rivets

1.2.1 Gather the Required Tools and Materials

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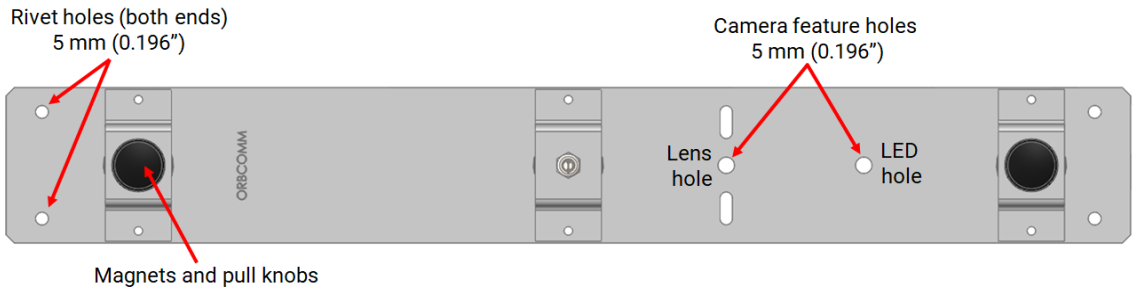
The following are required for this installation:

- CT 5300 with installed bracket (p/n CT5300-0120-H) includes:
  - rivets (two sizes provided)
  - alcohol based cleaner wipes
- Magnetic drill template (available from VIACHAIN, p/n ST101776)
- Duct tape (for non-ferrous assets)
- Rivet gun
- 5 mm (0.196") drill and drill stop. A 4.8mm (3/16") drill bit is an acceptable alternative.
- 57 mm (2¼") diameter hole saw with a 6.5 mm (¼") diameter pilot (Milwaukee Bi-Metal preferred)
- Marker
- Rags or towels
- Silicone sealant (gray or black recommended)
- Dry powder graphite lube
- Deburring tool
- Straight ladder (12' to 16' / 3.5 m to 5 m) with safety strap
- If installing VIACHAIN sensors, refer to the appropriate sensor installation guide.

1.2.2 Mark the Drill Hole Locations

CAUTION: If drilling on an asset (container or trailer) loaded with cargo, use drill stops.
- Attach the drill stop to the drill bit, the location of the stop on the bit will depend on the width of the asset wall.
- Use just enough force to slightly penetrate the asset wall. Going all the way through the asset wall may damage cargo.

1. Determine the appropriate location (vertically or horizontally) for the CT 5300. Avoid mounting over weld seams or rivets. All measurements are made to the camera **lens hole** on the drill template.



Note: When mounting in a vertical configuration the solar panel should face down for optimal RF performance.

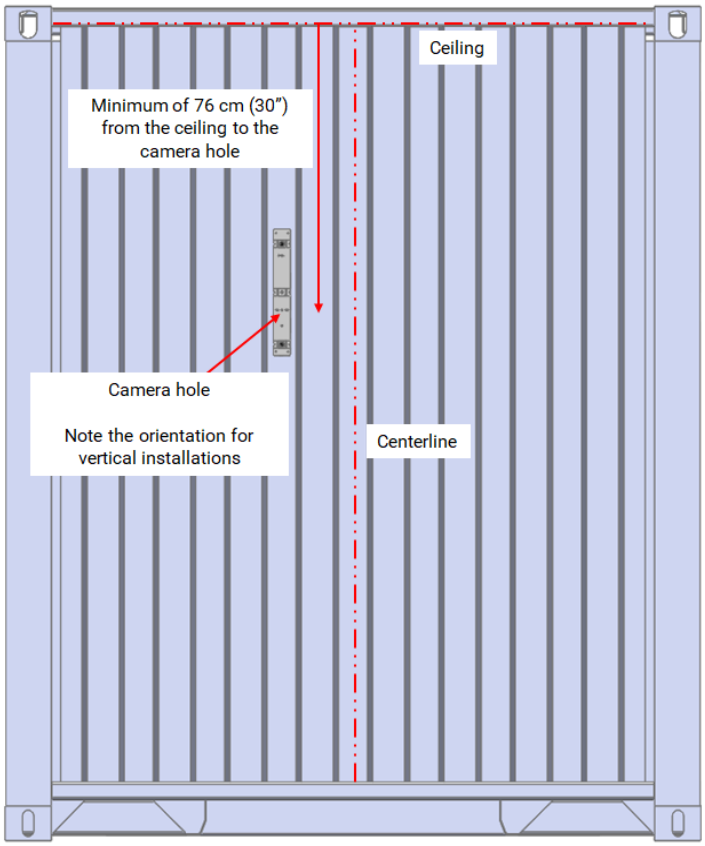
Note: When mounting is horizontal (on a container or thin-walled trailer), the solar panel can face either left or right.

Note: When mounting on containers with vertical corrugations, place the template one corrugate to the left or right if there is a weld seam along the center corrugate.

Note: Refer to the appropriate installation guide for any other sensor mounting instructions.

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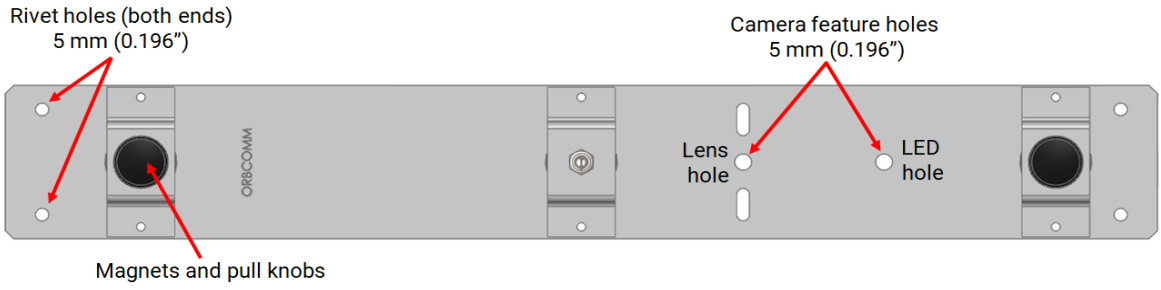
2. Place the template so that the camera **lens hole** is centered across the width of the asset (avoid weld seams) and as high up on the nose as possible but a minimum of 76 cm (30") from the ceiling.



3. Spot drill through the two (2) **camera feature holes** and the four (4) **rivet holes** on the template using a 5 mm (0.196") drill bit. A 4.8 mm (3/16") drill bit may be used if a 5 mm (0.196") drill bit is not available; however, the holes may need to be opened to fit the rivets.

CAUTION: DO NOT drill all the way through the container wall with the template in place.

CAUTION: DO NOT drill holes in placarded trailers that are loaded.



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1.2.3 Prepare the Holes

1. Remove the template.

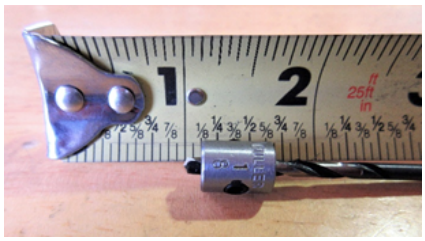
   CAUTION: DO NOT drill completely through the container or trailer with the template still attached. Only pilot holes should be drilled at this point and the template removed.

2. Secure a drill stop collar to the drill bit when drilling each of the 5 mm (0.196") holes.



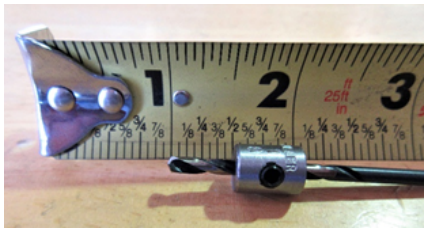
For Containers

Set the drill stop 3 mm (⅛") from the tip of the drill bit.



For Thin-Walled Trailer

Set the drill stop 12 mm (½") from the tip of the drill bit.



3. Drill the lower left drill hole location until the drill stop contacts the trailer or container surface.



4. Completely drill through the four (4) rivet holes to 5 mm (0.196") diameter.

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5. Use the 57 mm (2¼") diameter hole saw (recommended for all assets) with a 6.5 mm (¼") diameter pilot.

   For containers: The hole saw size can be smaller, if desired, with a minimum size of 45 mm (1¾") diameter hole saw, and a 6.5 mm (¼") diameter pilot.

6. Spray the teeth of the hole saw with dry powder graphite lube to reduce sparking or heat.

   CAUTION: If at any time sparks or excessive heat cannot be avoided, STOP the installation and call management.

   Note: DO NOT spray directly on the asset surface because this may leave residue.

7. Use the selected hole saw to cut through the predrilled camera feature holes.



8. Add another coat of dry powder graphite lube to the circular saw and then drill the second camera feature hole.
9. Remove excessively sharp edges, large / handing burrs or strips that are the result of using the hole saw, with a few quick passes of a deburring tool.

   Note: Only use edge deburring tools. Never use straight files or hand files as they will distort the shape of the hole.

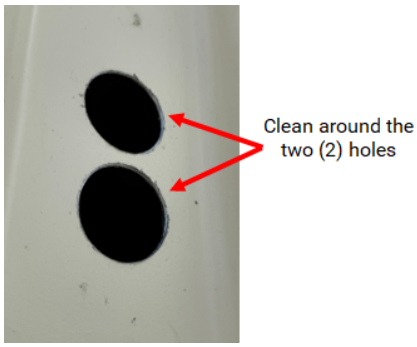


EXAMPLE OF A TYPICAL DEBURRING TOOL WITH A ROTARY STYLE HEAD

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1.2.4 Mount the Device

1. Use the supplied alcohol wipes to clean the surface of the asset around the two (2) large camera feature holes. This is where the external gasket on the mounting bracket seals against the asset.



2. Place the bracket on the asset in installation location. The camera features must line up with the two large holes in the asset.
3. Select the appropriate rivet size based on the asset. Two rivet sizes are included with the CT 5300 assembly.

   For containers, use the shorter rivets.

   For thin-wall trailers, use the longer rivets.
4. Insert one rivet through the bottom left hole of the bracket and one rivet through the top right hole of the bracket.

   CAUTION: This step is critical to align the camera features.

   CAUTION: It is very important to use the provided rivets as an alignment guide.

   Note: The following figure is the view from inside the container. Note the orientation in a vertical installation and the alignment of the camera features through the large holes.



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5. Assemble the rivets at both ends of the bracket, in four (4) places.



6. (optional) Add silicone sealant around the rivets if additional sealing is required.
7. Restart the CT 5300:
    - Use the ORBCOMM Installer app to restart the device.

      OR
    - Use the magnet to restart the device (APPENDIX C).

      Note: Magnet options are supported, but devices ship without a magnet.

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APPENDIX A CT 5300 WITH BRACKET DIMENSIONS

Figure 1: CT 5300 with Bracket Top View

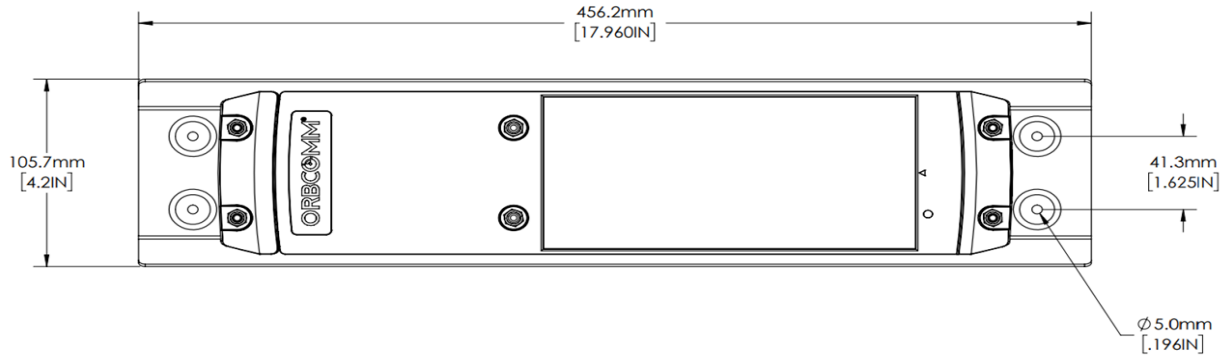


Figure 2: CT 5300 with Bracket Bottom View

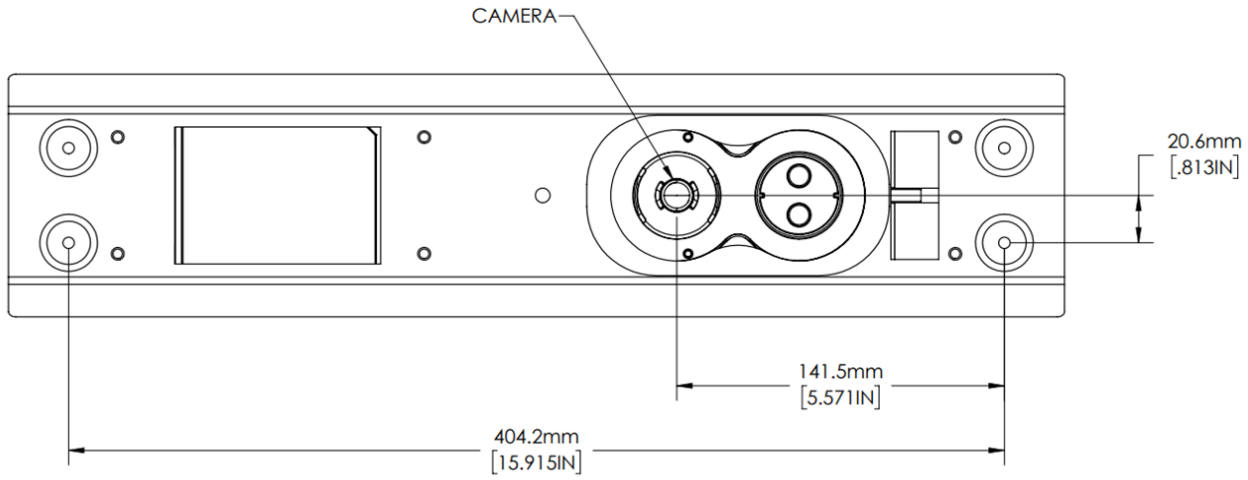
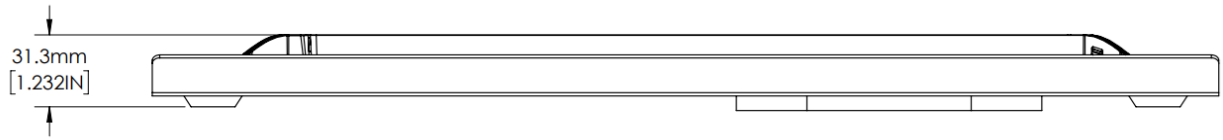


Figure 3: CT 5300 with Bracket Side View



Parameter	Description
Weight	
with rivet bracket	1.3 kg (3.0 lb)
without bracket	0.7 kg (1.5 lb)

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APPENDIX B CT 5300 LED INDICATORS

Green LED

The green LED is solid on for 15 seconds in the following cases:

- When the CT 5300 starts the application, including activation from shipping mode
- When I/O_1 becomes high

Blue LED

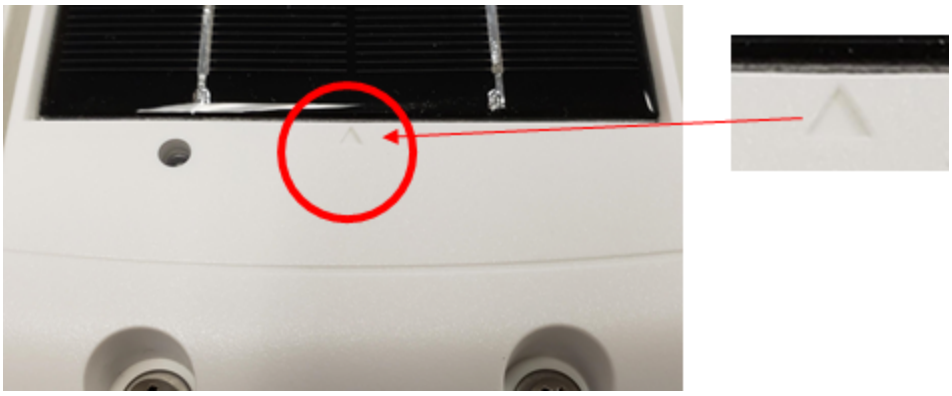
- The Blue LED turns on solid, while in the bootloader. The CT 5300 enters the bootloader after a reset.

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APPENDIX C CT 5300 MAGNET SWITCH ACTIONS

Note: Magnet options are supported, but devices ship without a magnet.

Figure 4: Correct Magnet Placement



Place a magnet[1] on or slightly raised above the arrow / triangle icon for	Action	LED activity when the magnet is removed	LED activity when the action is occurring
3 to 7 seconds and then remove	Executes a GNSS location update, and sends a from-terminal message if the asset is moving	Blinks green	Solid green for 15 seconds
55 to 65 seconds and then remove	Causes a restart	Blinks blue	Solid blue for 10 seconds

[1] Devices ship without magnets.

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